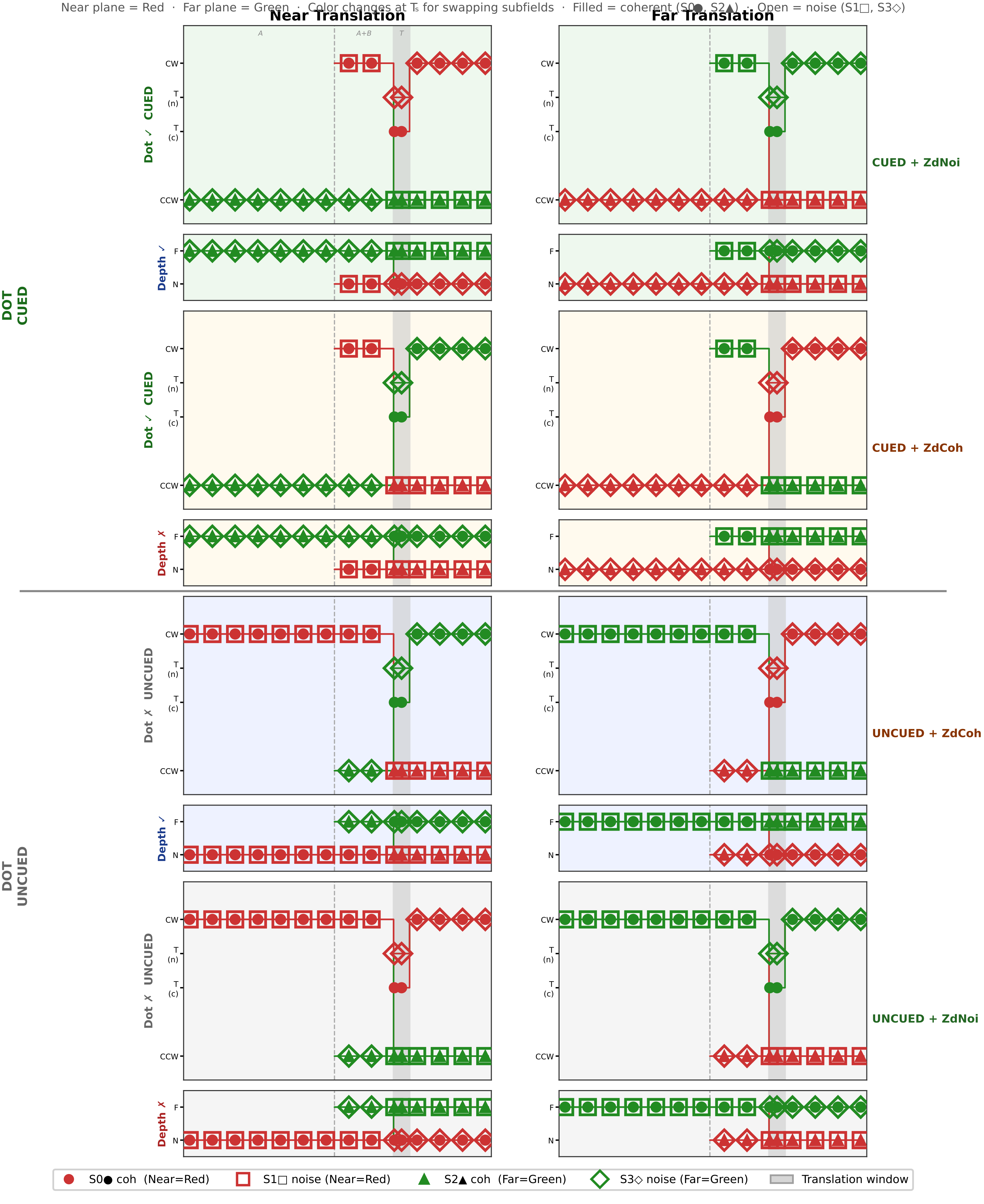
DepthColorLinked_005m — Design & Stimulus

Factor 1 — Dot Cueing (✓/✗): does the delayed-onset dot field translate?

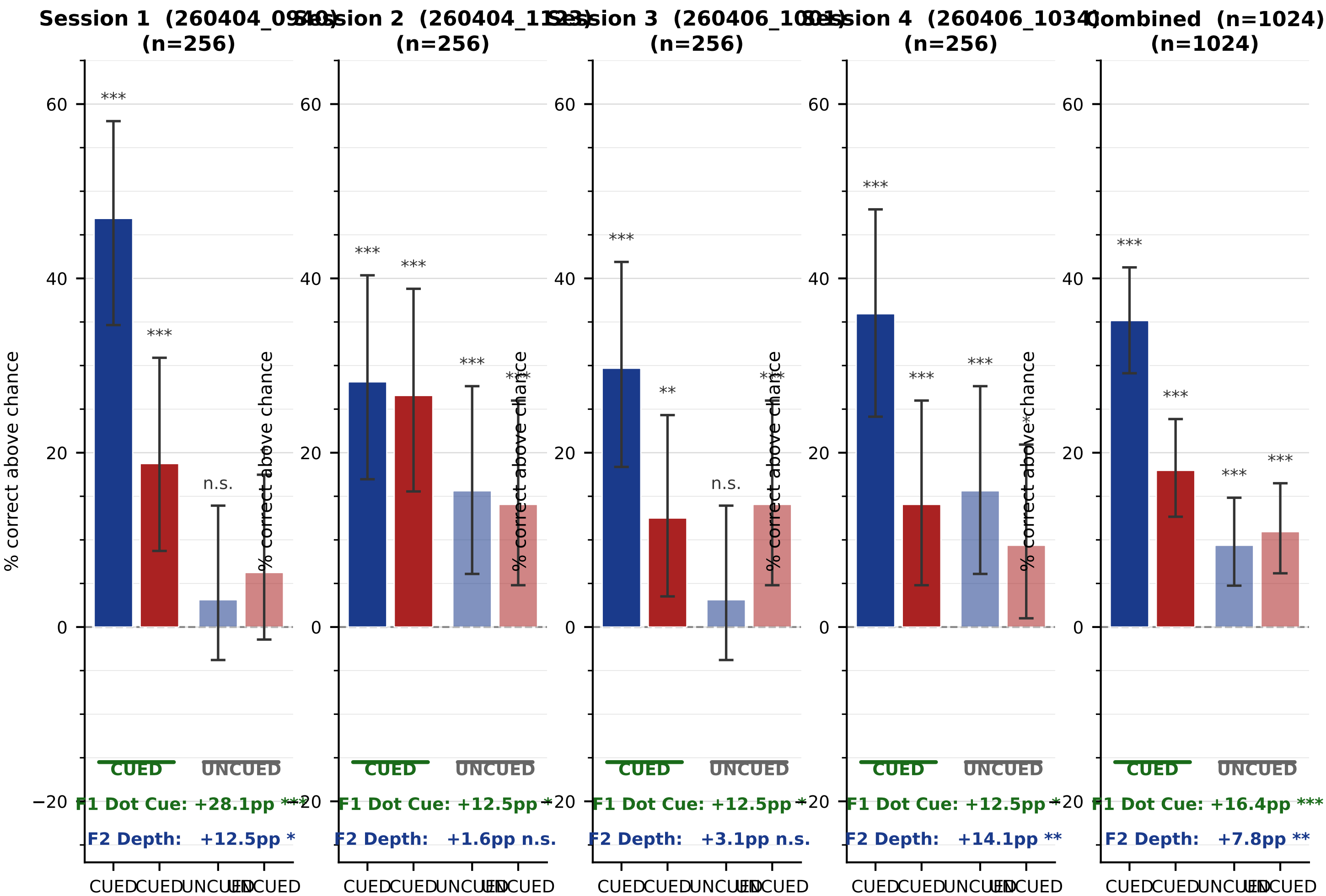
Factor 2 — Depth Cueing (✓/✗): does translation occur at the depth where the cued field appeared?

Near plane = Red · Far plane = Green · Color changes at T_s for swapping subfields · Filled = coherent (S0●, S2▲) · Open = noise (S1□, S3◇)



DepthColorLinked_005m — Results

Near=Red · Far=Green · depth sep = 0.05 m · 8-AFC · Bars = % correct above chance (12.5%) with 95% Wilson CI



Session comparison — main 2×2 cells (% correct above chance)

Condition	S1	S2	S3	S4	Combined	sig
CUED Depth✓ (ZdNoi)	+46.9pp (38/64)	+28.1pp (26/64)	+29.7pp (27/64)	+35.9pp (31/64)	+35.2pp (122/256)	***
CUED Depth✗ (ZdCoh)	+18.8pp (20/64)	+26.6pp (25/64)	+12.5pp (16/64)	+14.1pp (17/64)	+18.0pp (78/256)	***
UNCUED Depth✓ (ZdCoh)	+3.1pp (10/64)	+15.6pp (18/64)	+3.1pp (10/64)	+15.6pp (18/64)	+9.4pp (56/256)	***
UNCUED Depth✗ (ZdNoi)	+6.2pp (12/64)	+14.1pp (17/64)	+14.1pp (17/64)	+9.4pp (14/64)	+10.9pp (60/256)	***
F1 Dot Cueing (CUED−UNCUED)	**+28.1pp *****	**+12.5pp ***	**+12.5pp ***	**+12.5pp ***	**+16.4pp *****	
F2 Depth Cueing (Depth✓−Depth✗)	**+12.5pp ***	**+1.6pp n.s.**	**+3.1pp n.s.**	**+14.1pp ****	**+7.8pp ****	

DepthColorLinked_005m — Interpretation & Notes

Key Findings (4 sessions combined, n=1024)

F1 Dot Cueing: CUED substantially outperforms UNCUED (exact values page 2). This is the core temporal-onset cueing effect and replicates robustly across all three sessions.

F2 Depth Cueing: translation at the cued field's depth plane yields better performance than translation at the opposite plane (exact values page 2). Replicates the ZdA vs ZdB dissociation seen in DepthSwapCtrl (all-red).

Best condition: CUED+Depth✓ (ZdNoi). Translator keeps depth-plane identity and color — both cueing signals align. Dot cueing dominates: CUED+Depth✗ (ZdCoh) also exceeds UNCUED.

UNCUED performance: appears near-chance on average, but is heterogeneous by delayed-field depth (F3; see anomaly note below).

Between-Session Variability (4 sessions, n=256/session)

S1 (260404_0940): strong ZdNoi >> ZdCoh (CUED +47pp vs +19pp); UNCUED near chance. S2 (260404_1123): flatter — ZdNoi ≈ ZdCoh (~+28pp CUED); UNCUED elevated (+14–16pp). S3 (260406_1001): moderate ZdNoi >> ZdCoh (CUED +36pp vs +14pp); UNCUED ~+12pp. S4 (260406_1034): ZdNoi >> ZdCoh (CUED +36pp vs +14pp); F2 Depth Cueing strongest yet (**); F1 Dot Cueing modest (+12.5pp *). Observer reported low perceived performance.

F1 (Dot Cueing) is significant in all 4 sessions but varies in magnitude (S1=+28pp, S2–S4≈+12pp). F2 (Depth Cueing) significant in S1 (*) and S4 (**), near-null in S2–S3. Four-session combined n=256/cell provides reliable cell estimates.

Response bias: 90° and 270° headings are over-represented across sessions (observer motor/pointing bias plus possible vertical perceptual anisotropy). Not condition-specific.

UNCUED+Depth✗ (ZdNoi) Anomaly — Resolved

UNCUED+ZdNoi is nominally the "hardest" cell (Dot✗, Depth✗) yet performs above chance (~+11pp ***). This is not paradoxical — it is explained by F3 (translating-field depth).

In UNCUED+ZdNoi: the non-delayed (non-cued) field translates; ZdNoi leaves the non-coh companion in its original depth plane. The translating field is at the depth OPPOSITE to the delayed field (DFD). When DFD=Near, the translator is at Far (easy); when DFD=Far, translator is at Near (hard).

DFD=Near sub-cell: translator Far → +20.8pp *** (consistent with DepthParam Far >> Near). DFD=Far sub-cell: translator Near → +2.1pp n.s. (chance). The anomalous average (+11pp) is simply the average of an easy Far trial and a hard Near trial.

Implication: F2 Depth Cueing as currently defined is confounded with F3 (translating-field depth). A translating-field-depth reanalysis is needed to cleanly separate the two. The DepthParam parametric sweep (0.03–0.15 m) confirms Far translation is intrinsically easier, regardless of cueing condition.

Design Notes

Depth separation: 0.05 m at 2 m viewing distance (~1.4° disparity at aperture edge). Chosen from DepthParam parametric sweep; Far cueing robust at all depths tested (0.03–0.15 m). Near cueing noisier — may relate to monocular positional-shift confound that scales with eccentricity.

Color-depth confound: within this experiment, depth change and color change always co-occur (linkDepthColor=1). ZdCoh = depth+color change; ZdNoi = neither. Cannot separate the two within this design alone.

Cross-experiment comparison to DepthSwapCtrl (all-red, same depth): DepthSwapCtrl ZdA CUED = +12.5pp (bino, n=64); DepthColorLinked ZdA CUED (S1) = +18.8pp. Difference hints that adding color makes ZdCoh less disruptive, but sessions differ in baseline so interpretation is premature. Need matched n.

Factor Map

F1 Dot Cueing	CUED vs UNCUED — does the delayed-onset dot field translate?
F2 Depth Cueing	Depth✓ vs Depth✗ — does translation occur at the DFD during T?
Depth✓	= CUED+ZdNoi OR UNCUED+ZdCoh (translator at DFD during window)
Depth✗	= CUED+ZdCoh OR UNCUED+ZdNoi (translator at opp(DFD))
F3 TransDepth	Near vs Far — depth of translating field during window (post-hoc)
F4 TransColor	R vs G — color of translating field (perfectly correlated with F3)
F5 CuedColor	R vs G — color of delayed onset field (post-hoc)
F6 RotCfg	CW vs CCW rotation of delayed field (post-hoc, expected null)

Planned Next Steps

- Translating-field-depth reanalysis (F3): re-bin all cells by which depth plane (Near vs Far) translates, collapsing F1 and F2. Expect Far >> Near regardless of cueing.
- Cross-experiment comparison at matched n: DepthColorLinked (n=192/cell) vs DepthSwapCtrl all-red (n=64/cell bino) → isolate color contribution to ZdCoh disruption.
- Color-only swap experiment (linkDepthColor=0, no depth change): complete the 2×2 of depth-only × color-only to dissociate depth vs color contributions.
- Second observer and/or second observer sessions to assess generalizability.
- SOA experiment: vary delay between Field A onset and Field B onset to probe temporal window of cueing effect.